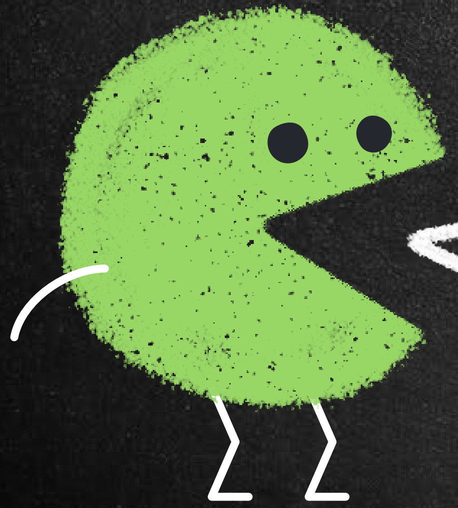
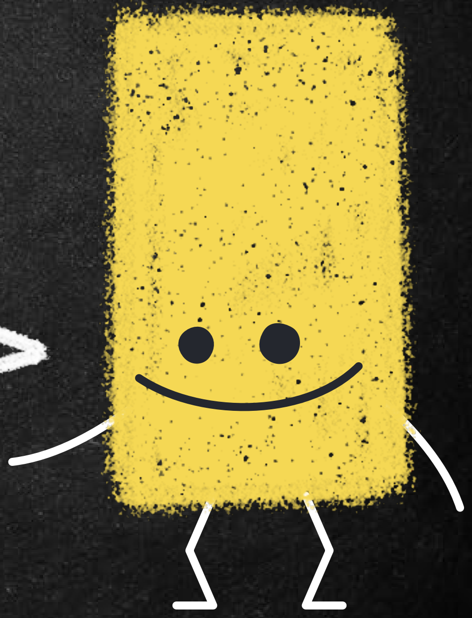
Digital Forensics
Part - 2

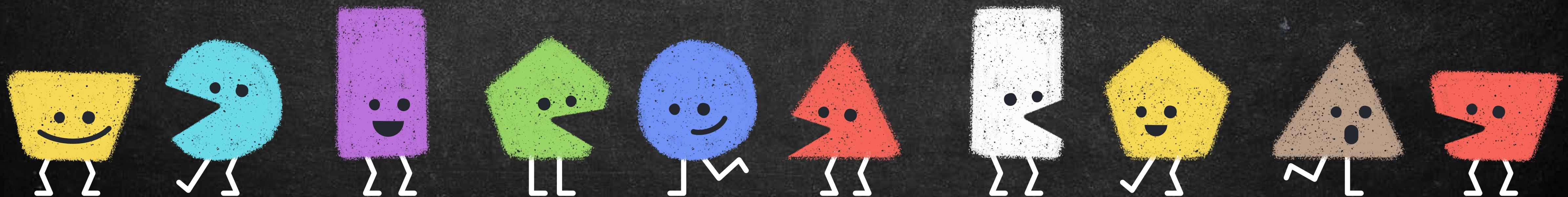




Hello!

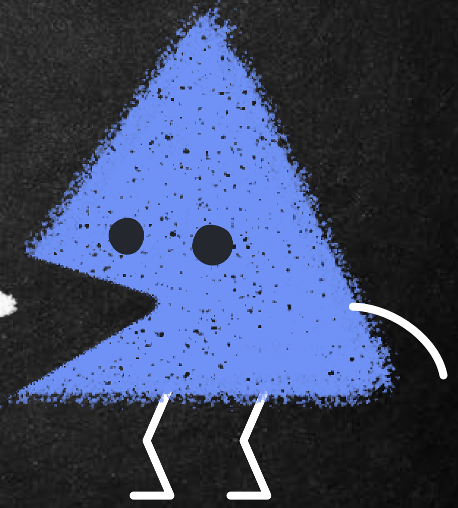
I am Sarang Rajvansh
Assistant Professor,
School of Cyber Security & Digital Forensics,
National Forensic Sciences university,
Gujarat Campus, Gandhinagar.

You can find me at sarang.rajvansh@nfsu.ac.in



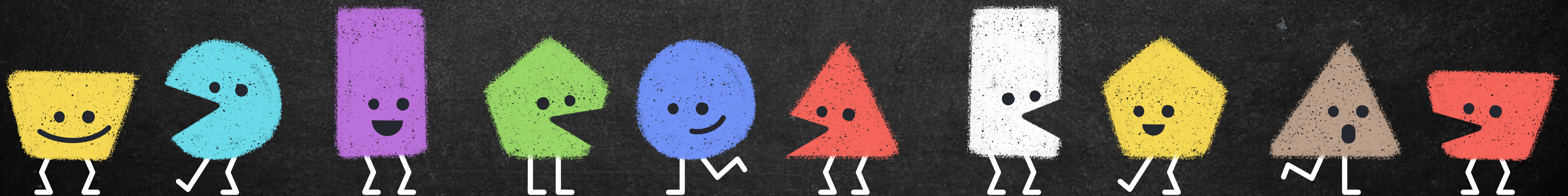
Basics

- → *The Boot Process*
  - *Kernel*
  - *BIOS*
  - *Bootstrapping*
- → *UEFI*
- → *MBR*
  - *Partition Table*
  - *Bootloader Code*
  - *Disk Signature*

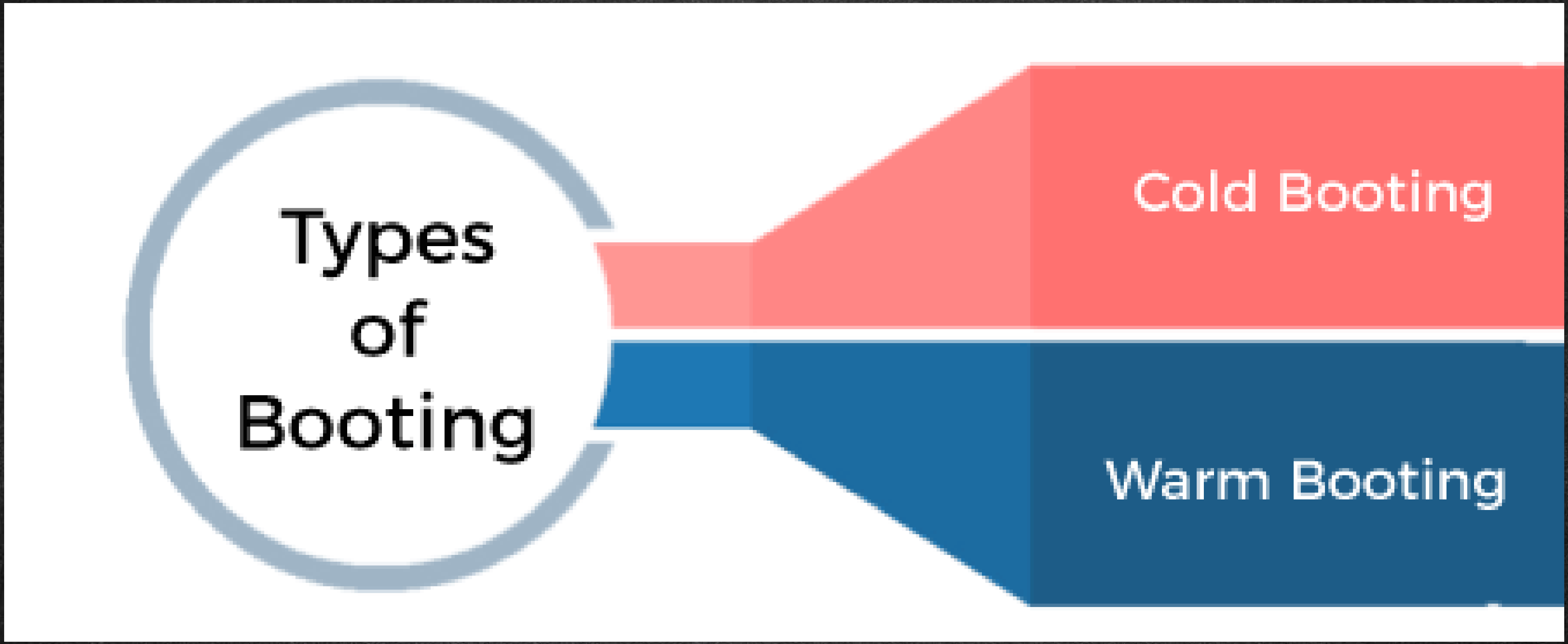


Booting Process

1. BIOS is loaded
2. Power-on self-test (POST) is completed
3. Operating system is loaded
4. System configuration is accomplished
5. System utilities are loaded
6. User is authenticated



Booting Process

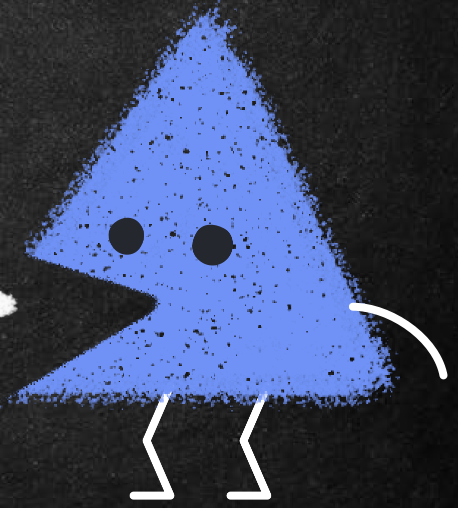


Kernel

- a kernel is a central component of an operating system that manages the system's resources and provides services to the other parts of the system.

- It is a low-level software layer that directly interacts with the computer's hardware.

- The kernel is responsible for managing system resources, such as memory, CPU time, and input/output (I/O) operations.

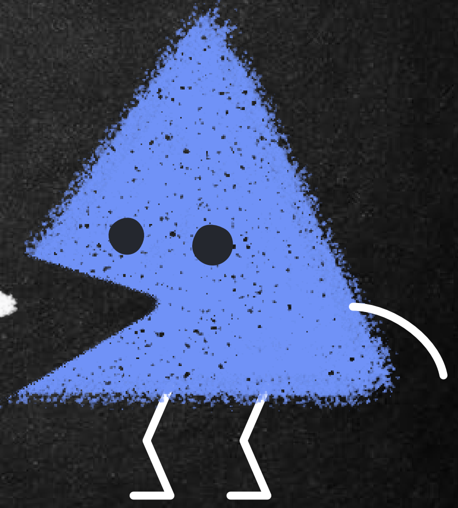


Kernel

→ It provides an interface for other software components to access these resources in a controlled and secure manner.

→ Additionally, the kernel also handles system calls, which are requests made by user programs to access the operating system's services.

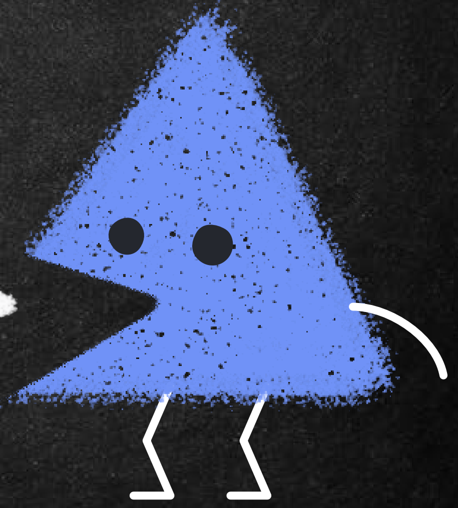
→ There are different types of kernels, such as monolithic kernels, microkernels, hybrid kernels, and exokernels.



Kernel

→ The choice of kernel design depends on the specific requirements of the operating system and the hardware it runs on.

→ The kernel is a critical component of the operating system, and any errors or vulnerabilities in the kernel can have serious consequences for the stability and security of the system.

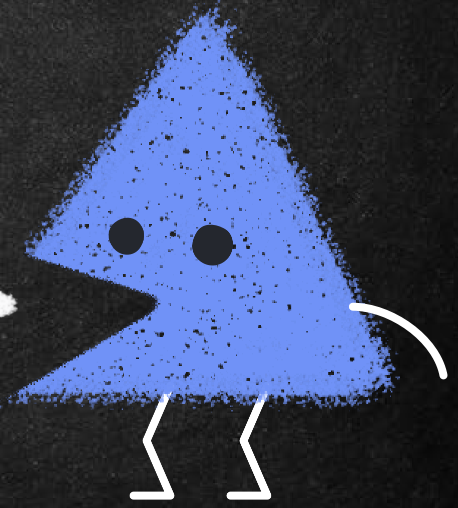


MBR

MBR

UEFI: Unified Extensible Firmware Interface

- → UEFI stands for "Unified Extensible Firmware Interface".
- → It is a specification for firmware, which is a type of software that provides low-level control of a computer's hardware.
- → UEFI replaces the older BIOS (Basic Input / Output System) firmware used in many computers.
- → UEFI provides a more modern interface for firmware, with support for larger hard drives, faster boot times, and more security features.

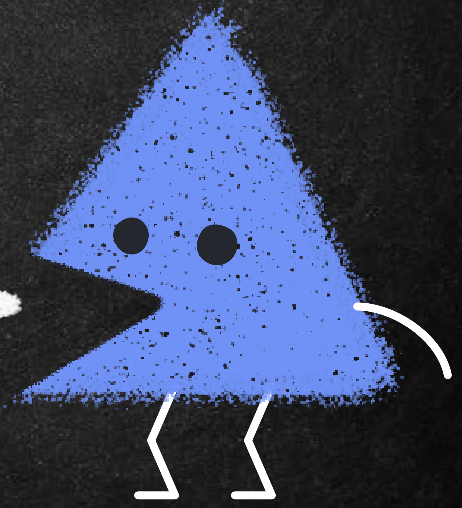


UEFI: Unified Extensible Firmware Interface

→ It also supports a wider range of hardware and software than the older BIOS firmware.

→ UEFI allows for pre-boot environments, which can run diagnostic tests and other utilities before the operating system is loaded.

→ One of the main advantages of UEFI is that it supports Secure Boot, a feature that helps protect against malware and other security threats by verifying that only trusted operating system software is loaded during the boot process.

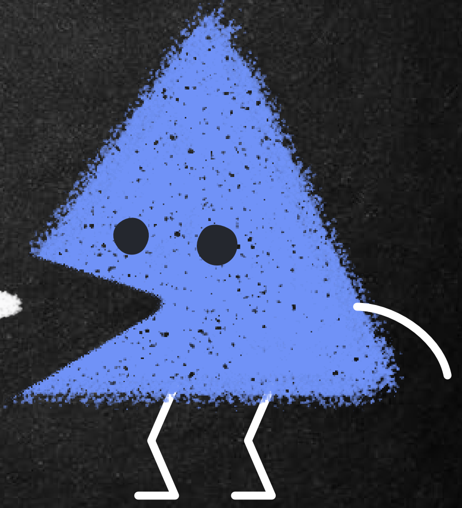


UEFI: Unified Extensible Firmware Interface

→ Overall, UEFI is a more advanced and flexible type of firmware that provides many benefits over the older BIOS firmware.

→ UEFI is made of

- UEFI firmware
- Boot Manager
- Boot Loader
- UEFI drivers
- UEFI Shell



MBR

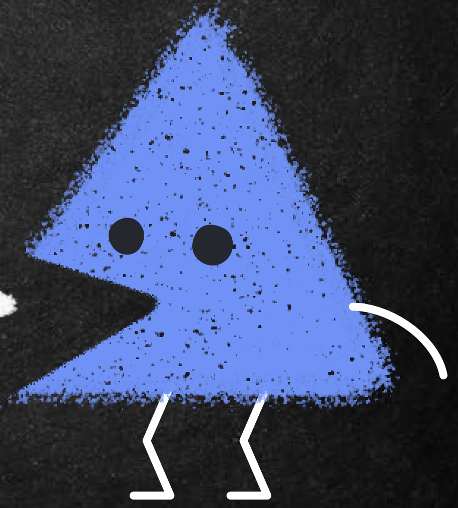
MBR

UEFI: Unified Extensible Firmware Interface

→ **UEFI firmware:** This is the core component of UEFI, which is responsible for initializing the hardware and starting the boot process.

→ The firmware contains a boot manager, which is responsible for finding and launching the operating system.

→ **Boot Manager:** This component of UEFI is responsible for selecting which device to boot from, such as a hard drive or USB drive. It presents a menu of boot options to the user, which can be selected using a keyboard or mouse.

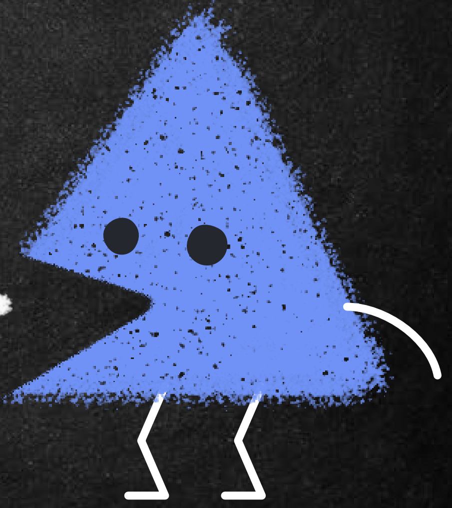


UEFI: Unified Extensible Firmware Interface

- → **Boot Loader:** Once the boot manager has identified which device to boot from, it loads the boot loader from that device.

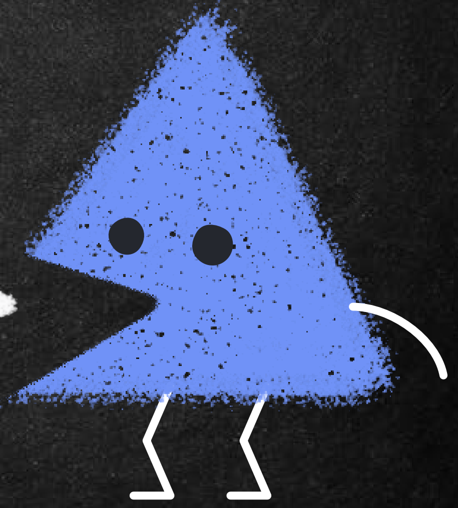
- → The boot loader is responsible for loading the operating system into memory and handing control over to it.

- → **UEFI drivers:** These are software components that provide support for hardware devices, such as network adapters, storage controllers, and graphics cards.



UEFI: Unified Extensible Firmware Interface

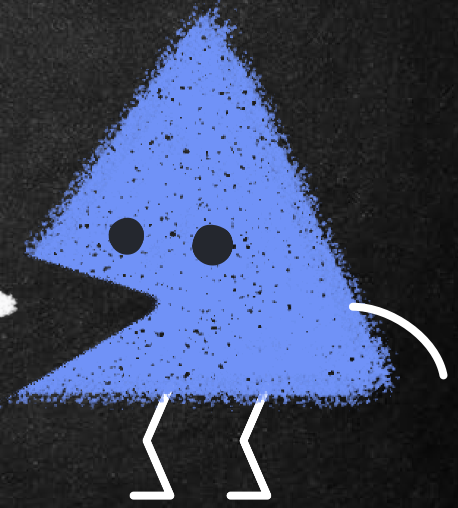
- UEFI drivers are stored in the firmware and can be updated separately from the operating system.
- **UEFI Shell:** This is a command-line interface that provides access to the UEFI firmware and its components.
- The shell can be used to perform diagnostic tests, modify firmware settings, and run utilities.



MBR: Master Boot Record

- → MBR is short for Master Boot Record. Typically, the MBR is the first sector on a startup drive (or other partitioned media). [Malwarebytes]

- → The MBR is the first sector on a hard disk and contains the partition table, which holds information on the number of partitions, their size and which one is active (i.e. which one contains the operating system used to boot the machine).[Kaspersky]



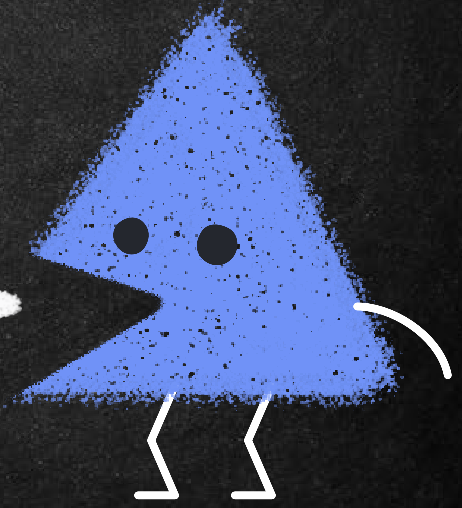
MBR: Master Boot Record

→ MBR is a small piece of code that is stored at the beginning of a hard disk drive or solid-state drive.

→ The MBR is essential for booting up an operating system, as it contains the necessary information for the computer's BIOS to locate and load the operating system.

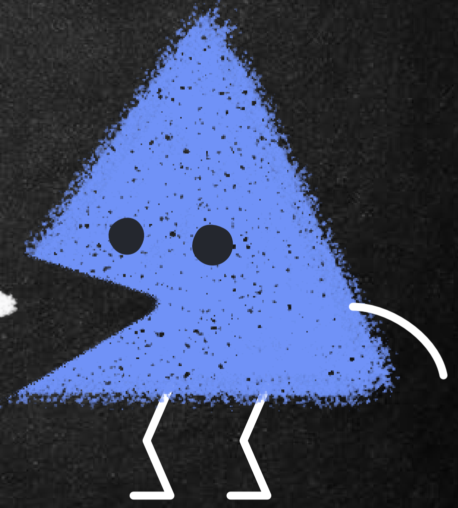
→ The MBR is made up of several components, including

- the boot loader code,
- partition table, and
- disk signature.



MBR: Master Boot Record

- Partition table: The partition table is a table that contains information about the partitions on the hard disk drive or solid-state drive.
- It includes information about the start and end sectors of each partition, as well as the partition type and file system used on the partition.
- The partition table is a data structure that describes the layout of the disk and how it is divided into separate partitions.

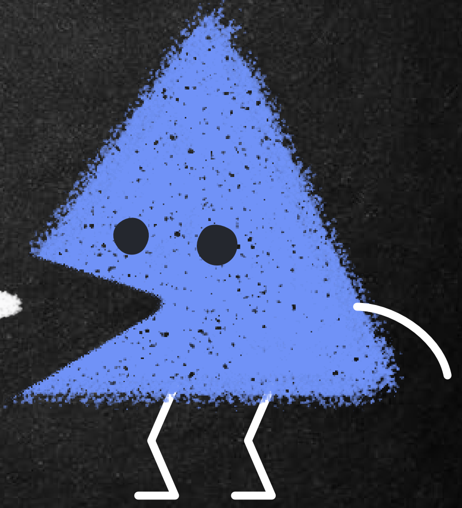


MBR: Master Boot Record

- → Each partition is a logical section of the disk that can be used to store files and data.

- → The partition table in the MBR can define up to four primary partitions, or three primary partitions and one extended partition.

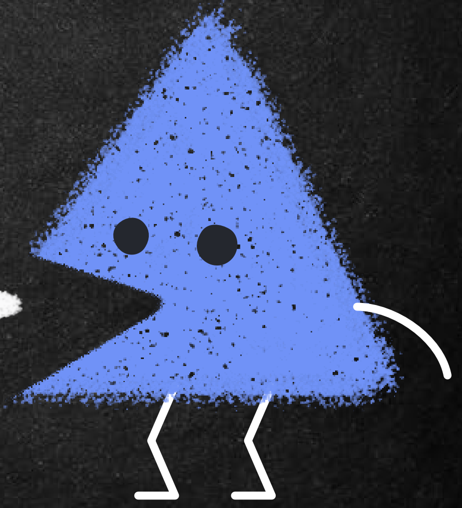
- → The extended partition can be divided into multiple logical partitions.



MBR: Master Boot Record

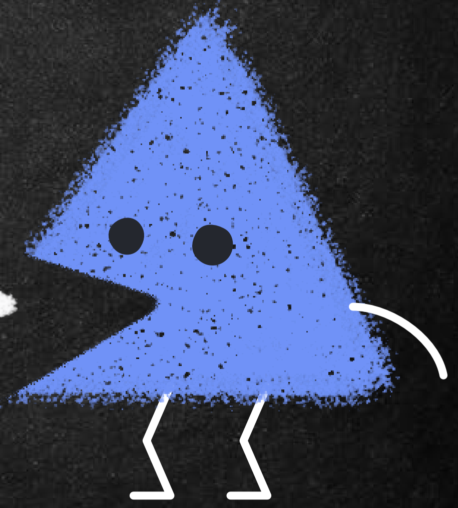
- → Boot loader code: This is the code that tells the computer where to find the operating system on the hard disk drive or solid-state drive.

- → It is usually located in the first 446 bytes of the MBR or in a separate boot partition.



MBR: Master Boot Record

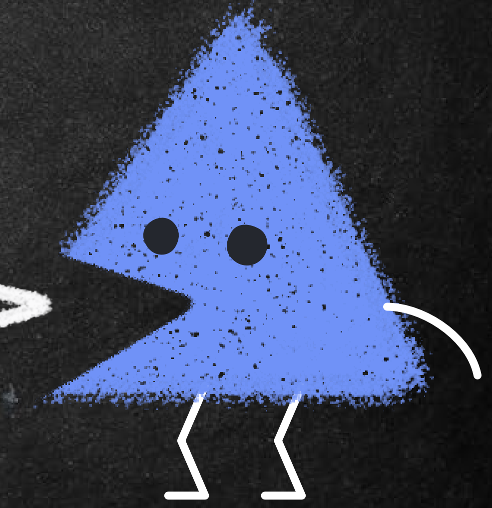
- → Disk signature: The disk signature is a unique identifier that is used to identify the disk.
- → It is stored in the last two bytes of the MBR.
- → Together, these components make up the contents of the MBR.
- → When the computer starts up, the BIOS reads the MBR to determine where to find the operating system, and then loads the boot loader code into memory to begin the boot process.



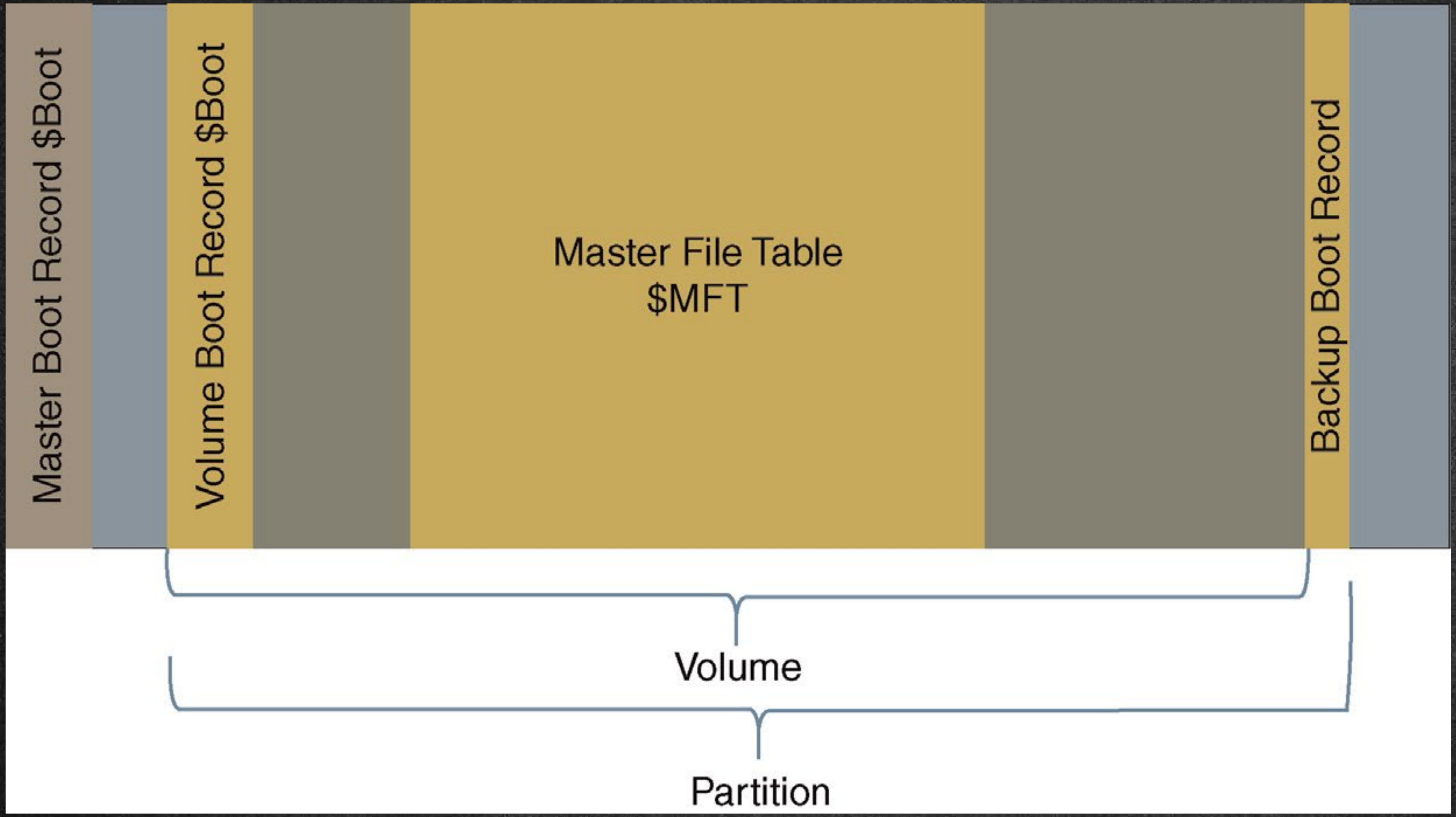
MBR: Master Boot Record

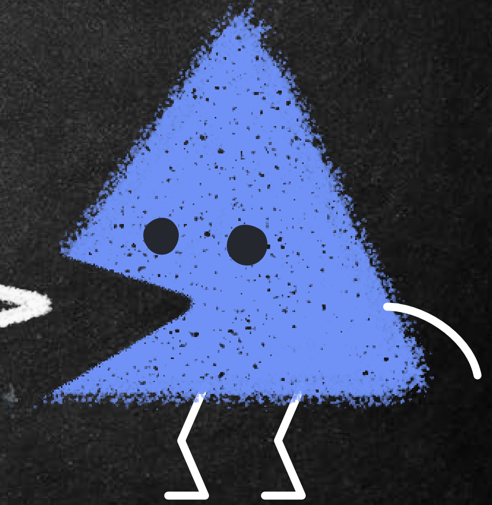
→ MBR is an older method of disk partitioning used with BIOS-based computers. [CentOS]

→ The sector-size of 512 bytes limits the disk size to 2TB and the number of primary partitions to 4, because of the maximum amount of information it can hold. [Malwarebytes]

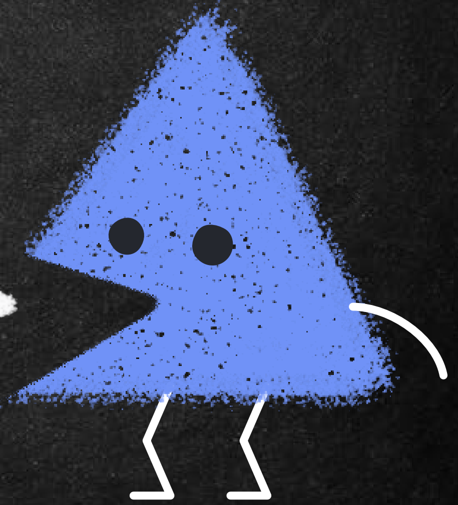


Logical File Structure

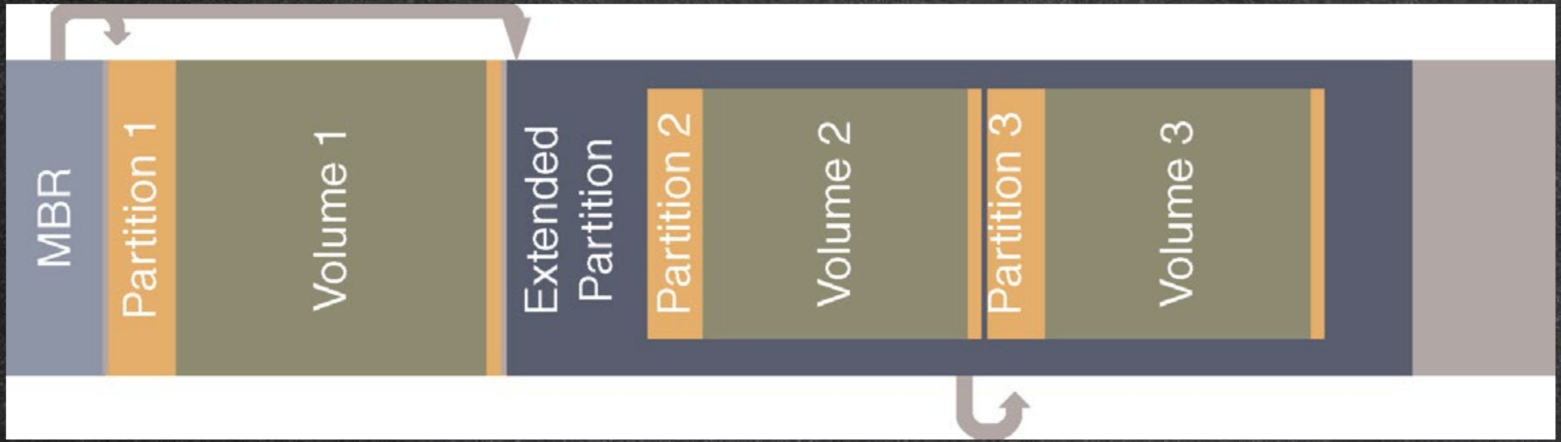




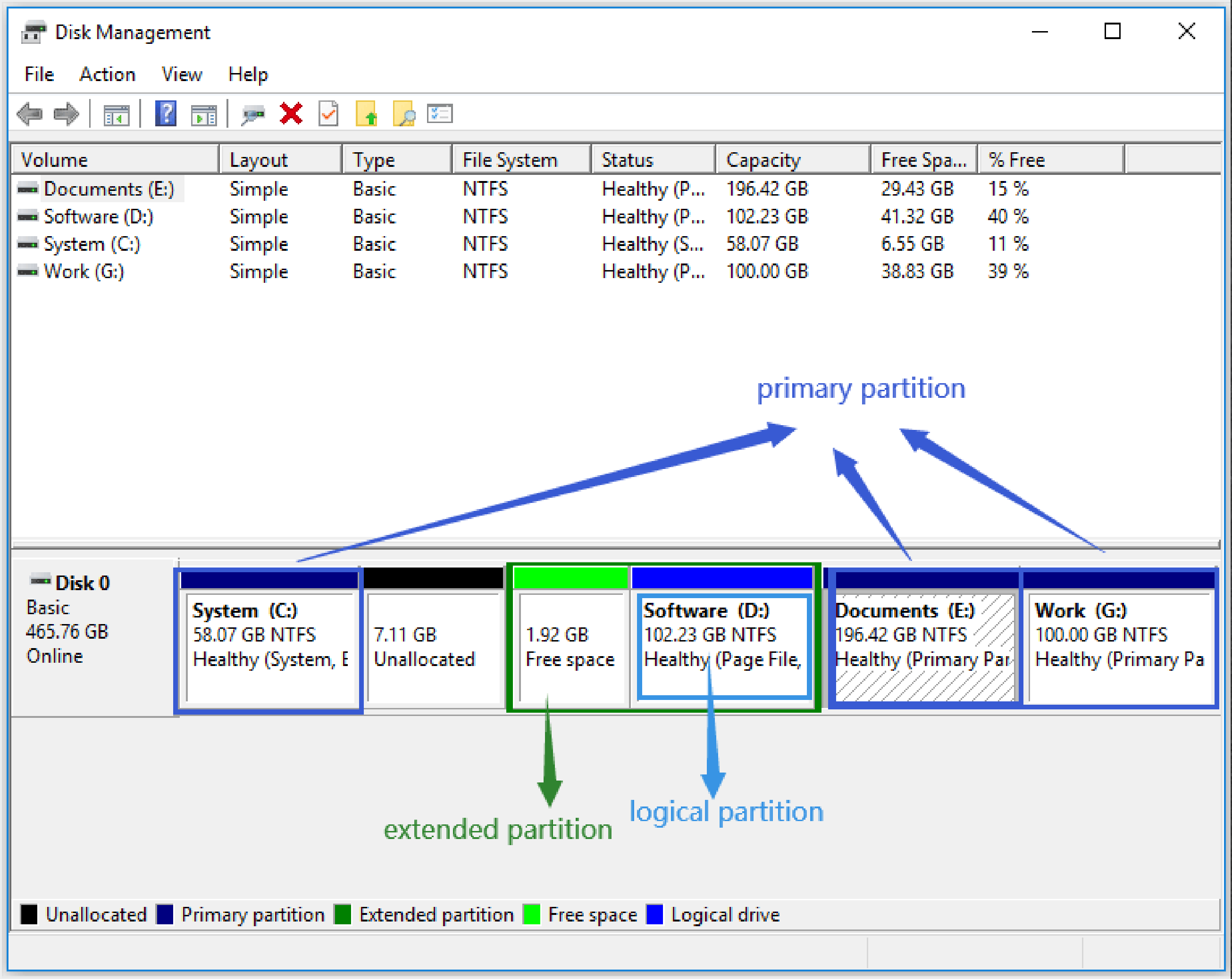
MBR Partition Structure



MBR Extended Partition Structure

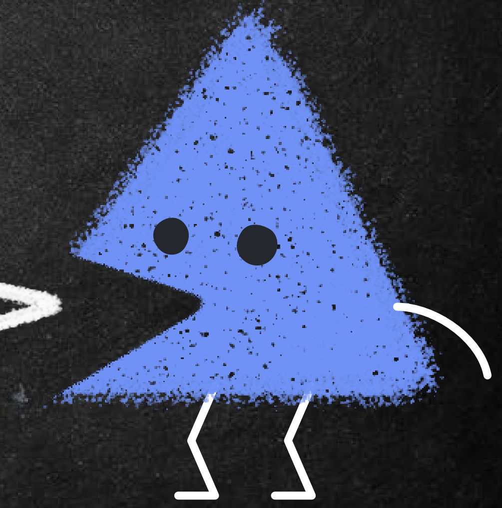


MBR Extended Partition Structure



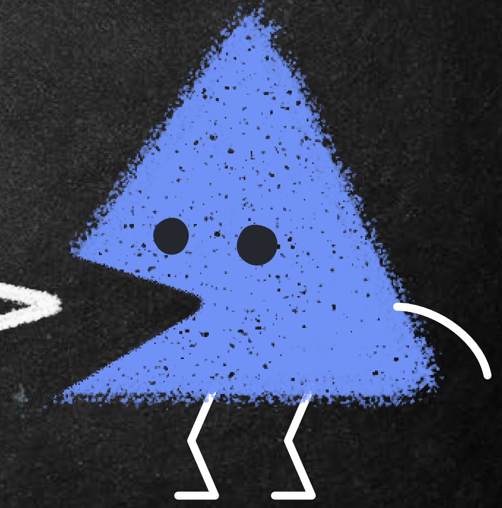
MBR Extended Partition Structure

Offset	Length	Contents
0x01BE	4	First partition's boot indicator and starting CHS address
0x01C2	4	Partition's starting sector (relative to start of disk)
0x01C6	4	Partition's size in sectors
0x01CA	4	Second partition's starting CHS address
0x01CE	4	Second partition's starting sector
0x01D2	4	Second partition's size
0x01D6	4	Third partition's starting CHS address
0x01DA	4	Third partition's starting sector
0x01DE	4	Third partition's size
0x01E2	4	Fourth partition's starting CHS address
0x01E6	4	Fourth partition's starting sector
0x01EA	4	Fourth partition's size
0x01EE	2	MBR signature (0x55AA)



Structure of a Modern Standard MBR

Address	Description		Size (bytes)
0x0000 (0)	Bootstrap code area (part 1)		218
0x00DA (218)	0x0000	*Disk timestamp*[2][b] (optional; Windows 95B/98/98SE/ME (MS-DOS 7.1–8.0). Alternatively, can serve as OEM loader signature with NEWLDR)	2
0x00DC (220)	Original physical drive (0x80–0xFF)		1
0x00DD (221)	Seconds (0–59)		1
0x00DE (222)	Minutes (0–59)		1
0x00DF (223)	Hours (0–23)		1
0x00E0 (224)	Bootstrap code area (part 2, code entry at 0x0000)		216 (or 222)
0x01B8 (440)	32-bit disk signature	*Disk signature* (optional; UEFI, Linux, Windows NT family and other OSes)	4
0x01BC (444)	0x0000 (0x5A5A if copy-protected)		2
0x01BE (446)	Partition entry №1	*Partition table* (for primary partitions)	16
0x01CE (462)	Partition entry №2		16
0x01DE (478)	Partition entry №3		16
0x01EE (494)	Partition entry №4		16
0x01FE (510)	0x55	*Boot signature*[a]	2
0x01FF (511)	0xAA		
		Total size: 218 + 6 + 216 + 6 + 4×16 + 2	512

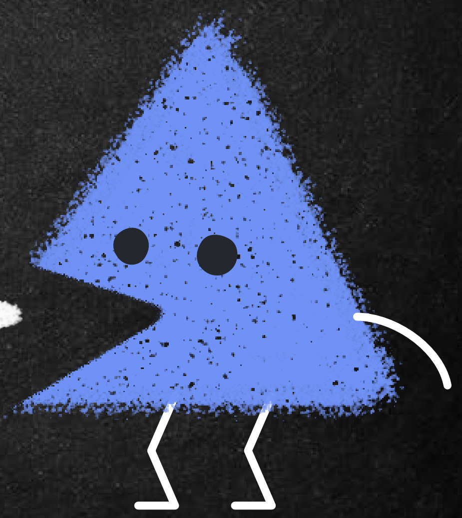


GPT: GUID Partition Table

→ GPT is a newer partitioning layout that is a part of the Unified Extensible Firmware Interface (UEFI). [Malwarebytes]

→ GPT disks use logical block addressing (LBA) and the partition layout is as follows:

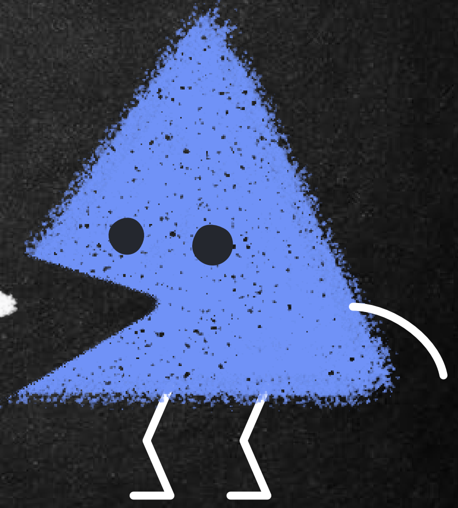
- To preserve backward compatibility with MBR disks, the first sector (LBA 0) of GPT is reserved for MBR data and it is called "protective MBR".
- The primary GPT header begins on the second logical block (LBA 1) of the device.



GPT: GUID Partition Table

- The header contains the disk GUID, the location of the primary partition table, the location of the secondary GPT header, and CRC32 checksums of itself and the primary partition table. It also specifies the number of partition entries of the table.

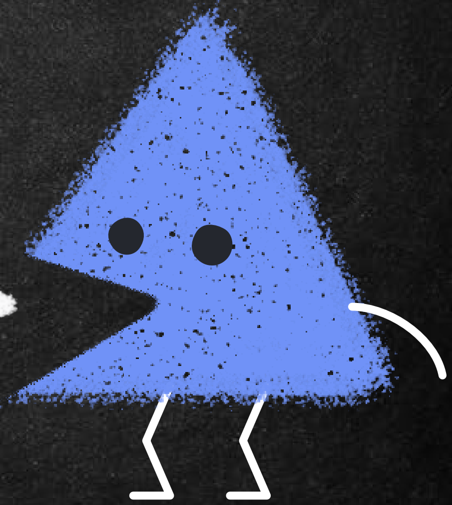
- The *primary GPT table* includes, by default, 128 partition entries, each with an entry size 128 bytes, its partition type GUID and unique partition GUID.



GPT: GUID Partition Table

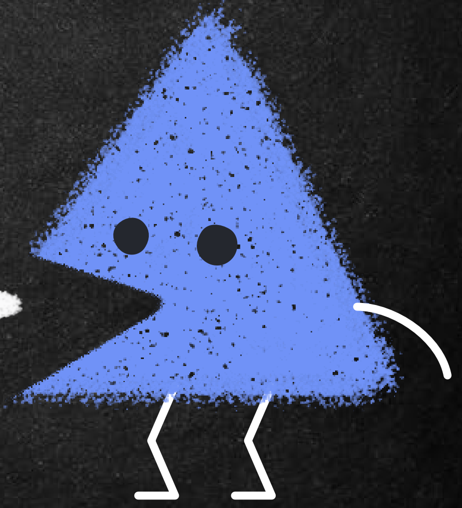
- The *secondary GPT table* is identical to the primary GPT table. It is used mainly as a backup table for recovery in case the primary partition table is corrupted.

- The *secondary GPT header* is located on the last logical sector of the disk and it can be used to recover GPT information in case the primary header is corrupted.



GPT: GUID Partition Table

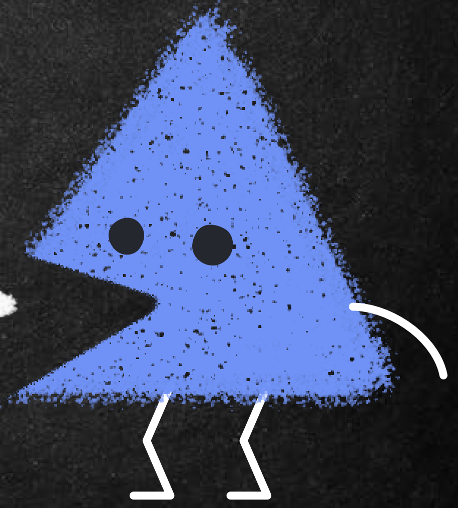
- It contains the disk GUID, the location of the secondary partition table and the primary GPT header, CRC32 checksums of itself and the secondary partition table, and the number of possible partition entries.



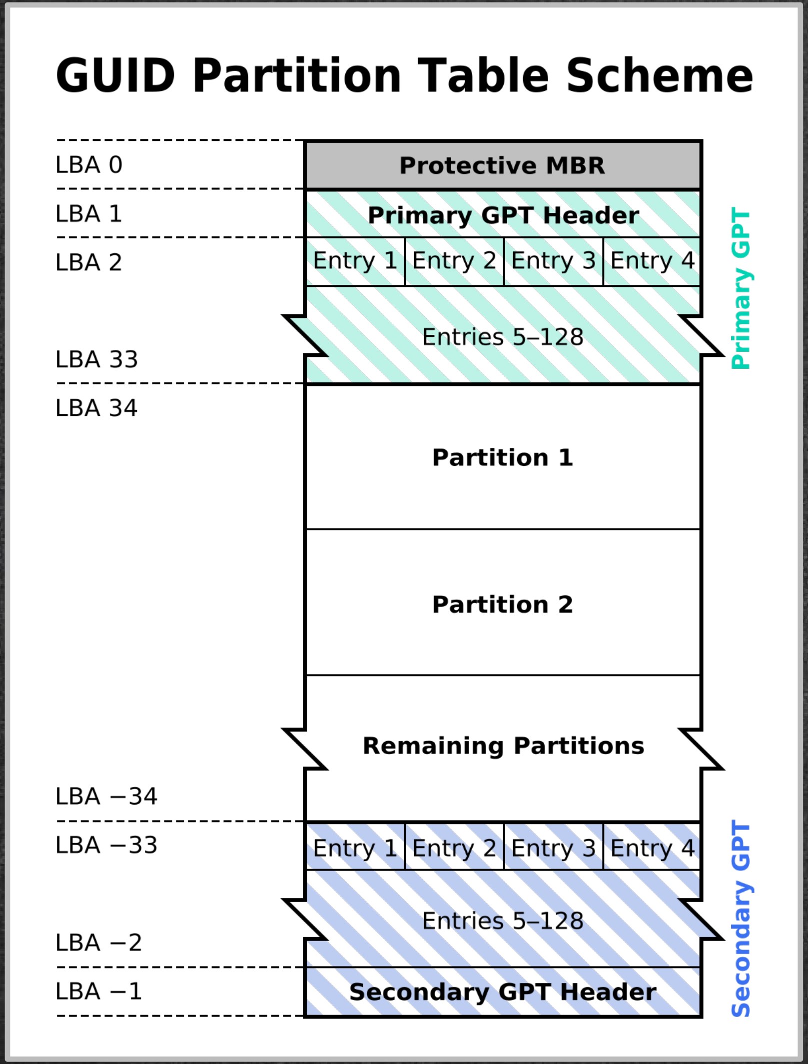
GPT: GUID Partition Table

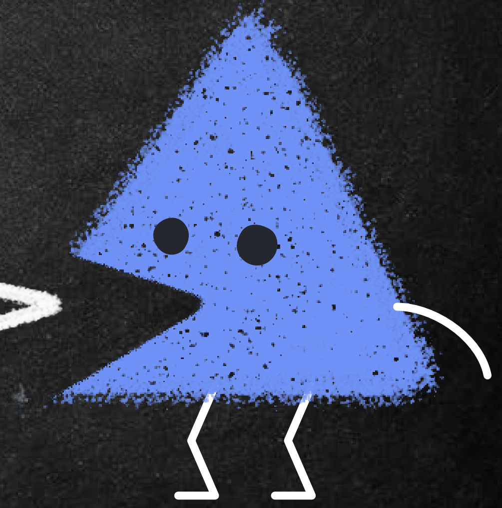
→ To address limitations of the MBR partitioning format, GPT schemes were created. Some advantages of this partition scheme include

- using GUIDs (global unique identifiers) to reference each partition;
- not limiting the number of partitions to a defined set of four;
- using 64 bit LBA for storing sector numbers, thereby increasing the theoretical maximum disk size; and
- maintaining backups and CRC32 checksums to detect and fix potential corruption.



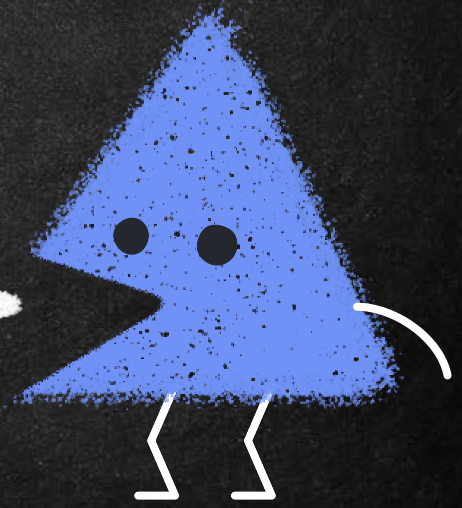
GPT: GUID Partition Table





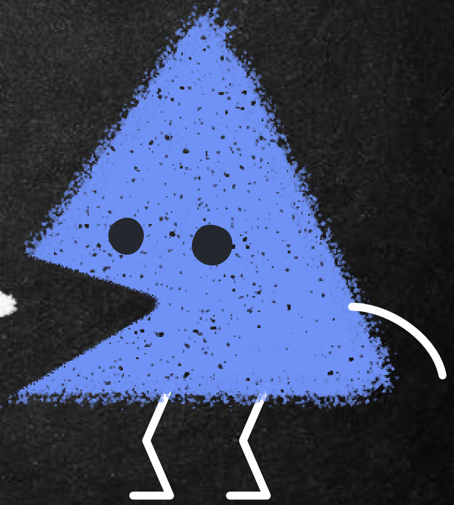
GPT: GUID Partition Table A

Offset	Length	Data
0x00	8bytes	Signature ("EFIPART")
0x08	4bytes	GPT revision level
0x0C	4bytes	GPT header size
0x10	4bytes	CRC32 of header
0x14	4bytes	Reserved
0x18	8bytes	LBA address of current header
0x20	8bytes	LBA address of backup header
0x28	8bytes	First addressable LBA for partitions
0x30	8bytes	Last addressable LBA for partitions
0x38	16bytes	Disk GUID
0x48	8bytes	Starting LBA of partition entry array
0x50	4bytes	Number of partition entries in the array
0x54	4bytes	Size of a single partition entry
0x58	4bytes	CRC32 of partition array
0x5C	420bytes	Reserved



GPT: GUID Partition Table B

Offset	Length	Data
0x00	16bytes	Partition type GUID
x10	16bytes	Unique partition GUID
x20	8bytes	First addressable LBA
x28	8bytes	Last addressable LBA
x30	8bytes	Attribute flags
x38	72bytes	Partition name

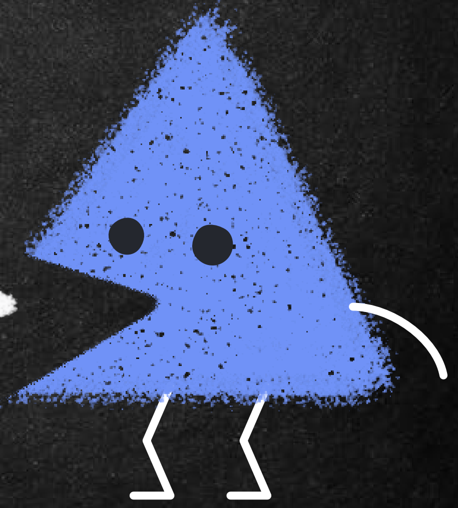


GPT: GUID Partition Table

- → The GPT partition scheme reserves absolute sector zero and utilizes the same offset as the MBR uses to describe a partition.

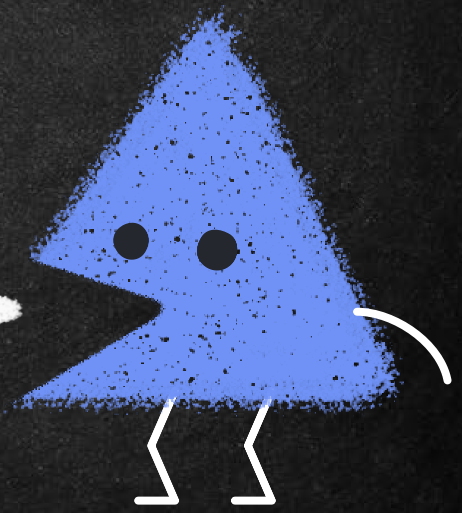
- → This first entry describes the entire disk up to 2 TiB (the maximum size definable in an MBR). The next sector contains the GPT header and defines the disk contents, as shown in Table A and Table B.

- → Contents include the GUID for the disk, partition table location, maximum number of partitions, and location of the backup header.



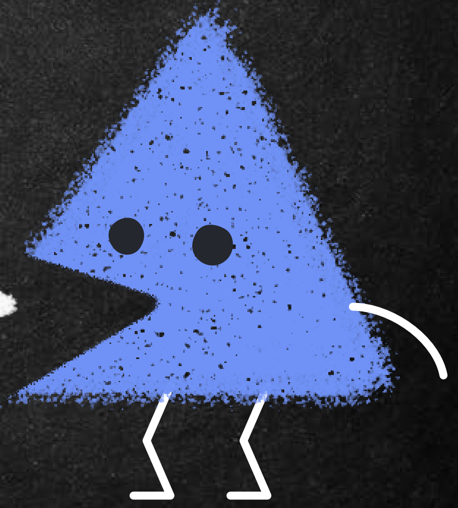
GPT: GUID Partition Table

MBR	GPT
Master Boot Record	GUID Partition Table
A legacy partitioning scheme used to boot computer systems and define the partition layout of storage devices	A newer partitioning scheme that offers more advanced features and greater flexibility in defining partition layouts of storage devices
Limited to 2TB for hard disks and 32GB for solid-state drives	Supports up to 9.4 zettabytes (ZB)
Supports up to four primary partitions or three primary partitions and one extended partition	Supports up to 128 partitions
Uses a traditional partition table layout, where partition information is stored in a single entry in the MBR	Uses a partition table layout with multiple entries that are distributed across the disk, allowing for redundancy and fault tolerance



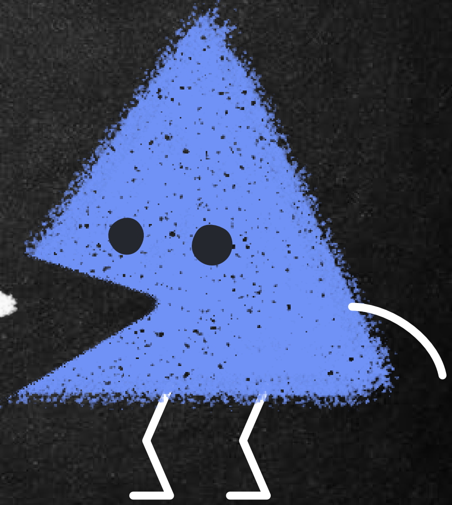
GPT: GUID Partition Table

MBR	GPT
Widely supported by most operating systems and boot loaders	Requires UEFI firmware and is not fully supported by some older operating systems and boot loaders
Vulnerable to malware attacks and disk corruption due to its reliance on the boot sector	Offers enhanced security features, such as support for secure boot and partition-level encryption
Does not offer built-in backup and recovery tools	Offers a built-in backup and recovery mechanism through its redundant partition table layout and CRC32 checksums
Commonly used on older systems or on smaller storage devices where advanced features are not required	Preferred for newer systems and larger storage devices due to its advanced features and greater flexibility in defining partition layouts



Additional Read

- → https://www.hp.com/us-en/shop/tech-takes/what-is-uefi
- → https://www.dell.com/support/kbdoc/en-in/000133480/uefi-and-secure-boot-faqs
- → https://learn.microsoft.com/en-us/windows-hardware/design/device-experiences/oem-uefi



References

- Digital Forensics by André Årnes
- Computer Systems: Digital Design, Fundamentals of Computer Architecture and Assembly Language by Ata Elahi
- Practical Forensic Imaging: Securing Digital Evidence with Linux Tools by Bruce Nikkel
- A Practical Guide to Digital Forensics Investigations 2nd Edition by Darren R. Hayes